

Thermodynamics

Grau en Enginyeria Matemàtica i Física

Joan Rosell

Departament d'Enginyeria Química
Universitat Rovira i Virgili, Tarragona

joan.rosell@urv.cat

Second Quadrimester 2024

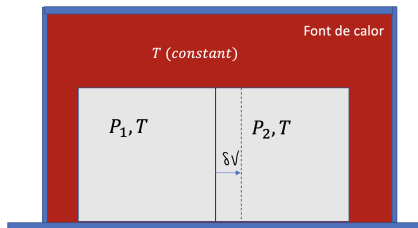
5. Material equilibrium

5.1. Material equilibrium

Equilibrium and stability

The minima conditions studied in our introduction of thermodynamic potentials allow us to establish conditions of equilibrium and thermodynamic stability, which will be fundamental to understand phase equilibria (but also reaction equilibria).

To explain the concept, consider an isothermal system at constant volume internally separated by a diathermal piston, as indicated in the figure.



5.1. Material equilibrium

Equilibrium and stability

To study the equilibrium and stability of the system under the piston displacement, we take as starting point the Helmholtz free energy, because in a system at constant T and V , it satisfies $\Delta F < W^{useful}$. Here, the work we extract is zero, so any *spontaneous* process taking place will lead to the minimum of F .

Let's assume that the system is at this minimum and we want to study how the free energy changes when we make small changes in the volumes of subsystems 1 and 2. If system 1 increases its volume by δV , then system 2 will decrease its volume by the same amount. Then, if we write $F = F_1(T, V_1) + F_2(T, V_2)$ and

$$\delta F = F_1(T, V_1 + \delta V) + F_2(T, V_2 - \delta V) - (F_1(T, V_1) + F_2(T, V_2))$$

considering a Taylor series expansion,

(next slide)

5.1. Material equilibrium

Equilibrium and stability

Since δV is arbitrary, the only possibility for F to be minimal is that the linear term is identically zero; thus,

$$P_1 = P_2$$

As we had announced, the principle of minimum F leads to the fact that the pressures in each compartment must be equal for the system to be in equilibrium. We knew this intuitively because we have a mechanical understanding of pressure and know that mechanically the forces must balance. The established procedure, however, is applicable to situations for which we lack an intuition from other disciplines.

Our second conclusion is that

$$-\left. \frac{\partial P}{\partial V} \right|_T > 0$$

The inverse of this quantity (normalized) is called isothermal compressibility, and is assigned

the symbol κ :

$$\kappa \equiv -\frac{1}{V} \left. \frac{\partial V}{\partial P} \right|_T > 0$$

If we isothermally increase the pressure on a stable system, its volume decreases.

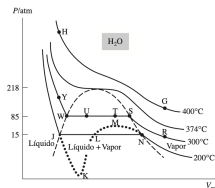
5.1. Material equilibrium

Phase Equilibrium

Let us now consider a non-ideal gas, represented by a van der Waals gas, which satisfies the equation of state

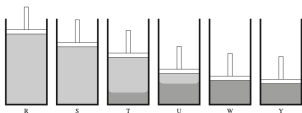
$$P = \frac{RT}{v - b} - a \left(\frac{1}{v} \right)^2$$

where $v = V/N$ is the molar volume. If we represent the isotherms, they have the form shown in the figure.



The regions where $\kappa > 0$ is the region where the derivative of P with respect to v is negative, and the system is thermodynamically stable.

There is a region, however, where $\kappa < 0$ (region between point K and point M for the dotted isotherm J-N). In this region the system is physically unstable and cannot exist as a single phase.



5.1. Material equilibrium

Phase Equilibria

The limits of the stability region define the curve named *spinodal*. The points on this curve satisfy the equation

$$\left. \frac{\partial P}{\partial v} \right|_T = 0$$

For example, for the van der Waals gas case, these points satisfy a third degree equation,

$$\frac{RT}{(v-b)^2} + 2a\frac{1}{v^3} = 0$$

The critical point is located on the critical isotherm, and is an inflection point with horizontal slope. In other words,

$$\left. \frac{\partial P}{\partial v} \right|_T = 0 \text{ and also } \left. \frac{\partial^2 P}{\partial v^2} \right|_T = 0$$

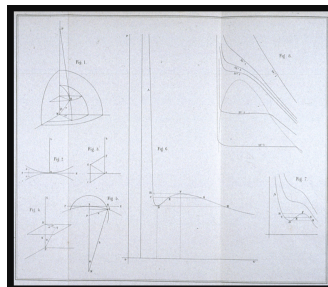
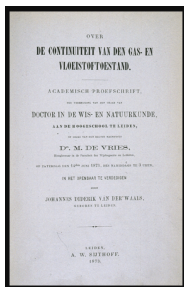
By solving these equations [6], we find that

$$v_c = 3b; \quad P_c = \frac{a}{27b^2}; \quad T_c = \frac{8a}{27Rb}$$

5.1. Material equilibrium

Phase Equilibria

Van der Waals [4] already realized that the shape of his isotherms did not correspond to what is observed experimentally in the two-phase region (solid line J-L-N).



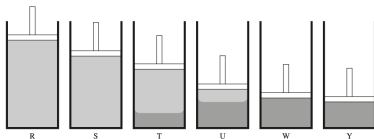
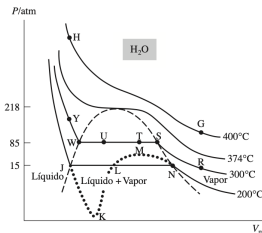
Experimentally, a flat line was found in equilibrium, between the molar volume of the liquid and that of the gas (**Anderson**). These points lie beyond the line that separates the thermodynamically stable from the unstable region, but the above analysis is unable to find those points.

5.1. Material equilibrium

Phase Equilibria

Condition of material equilibrium

The solution to this problem is due to **Maxwell** [5]. Consider again the figure:



Now consider the *isobaric* path in the figure which allows one to go from gas to liquid in equilibrium simply by doing work on the system at constant P and T . These conditions suggest that the relevant thermodynamic potential for this process is not F but G . In addition, the situation is no longer an impermeable piston that separates the phases but a permeable one (the liquid-vapor interface). Then, we can write

$$G(T, P, N^l, N^g) = G_l(T, P, N^l) + G_g(T, P, N^g)$$

5.1. Material equilibrium

Phase Equilibria

Condition of material equilibrium

Since the system is in equilibrium (at constant T and P), G reaches a minimum, so $dG = 0$. If we assume that the two phases can virtually exchange δn moles while keeping the total number of moles constant,

$$0 = dG = dG_l + dG_g = G_l(T, P, N^l + \delta n) - G_l(T, P, N^l) + G_g(T, P, N^g - \delta n) - G_g(T, P, N^g) = \left. \frac{\partial G_l}{\partial N^l} \right|_{T,P} (\delta n) + \left. \frac{\partial G_g}{\partial N^g} \right|_{T,P} (-\delta n) = (\mu^l - \mu^g) \delta n$$

Since δn is arbitrary, the only possibility for $(\mu^l - \mu^g)\delta n$ to be 0 is that

$$\mu^l(T, P) = \mu^g(T, P) \quad (1)$$

where, since $\mu(T, P, N^i)$ is intensive, it is only a function of T and P (for a single component).

5.1. Material equilibrium

Phase Equilibria

Condition of material equilibrium

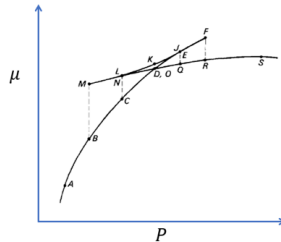
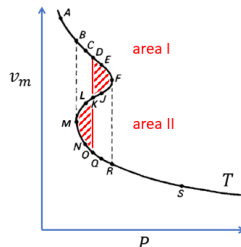
Now we can follow the evolution along the isotherm, as it crosses the region of phase coexistence [2]. To do so, we must use the Gibbs-Duhem equation to relate the changes in v_m at constant T to the changes in μ :

$$d\mu = -s dT + v dP$$

$$\mu = \int v dP + \phi(T)$$

where $\phi(T)$ is a 'constant of integration'. Therefore, along an isotherm, we would find that the chemical potential evolves qualitatively as shown in the figure. The system always 'chooses' the minimum- G branches (recall that $G = \mu N$). These cross at D and O, where: $\mu_D = \mu_O$. This corresponds to:

$$\text{area I} = \text{area II}$$

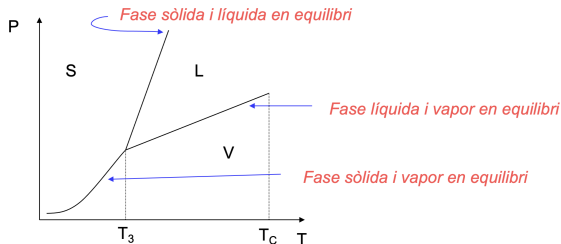


5.1. Material equilibrium

Phase Equilibria

Multiple phases in equilibrium for a single component ($n = 1$)

Let us consider the simple system with the following phase diagram (P - T diagram showing the different phases and their *coexistence lines*).



- ▶ The liquid-vapor line ends at a **critical point**.
- ▶ The solid-liquid line has no end.
- ▶ The three lines intersect at the so-called **triple point**, at which the three phases coexist.

Each line satisfies $\mu^\alpha(T, P) = \mu^\beta(T, P)$. At the triple point, the chemical potential is the same in each of the three phases:

$$\mu^S(T, P) = \mu^L(T, P) = \mu^V(T, P)$$

These two equations determine the values of $T = T_3$ and $P = P_3$.

5.1. Material equilibrium

Phase Equilibria

Multicomponent systems

Now, suppose we have $i = 1, 2, \dots, n$ components in a mixture for which two phases, α and β (liquid, solid, or gas, in simple systems; or concentrated solution, dilute solution, etc., in complex systems) coexist in material equilibrium at given T and P . The following system of equations is satisfied:

$$\begin{aligned}\mu_1^\alpha(T, P, N_1^\alpha, N_2^\alpha, \dots, N_n^\alpha) &= \mu_1^\beta(T, P, N_1^\beta, N_2^\beta, \dots, N_n^\beta) \\ \mu_2^\alpha(T, P, N_1^\alpha, N_2^\alpha, \dots, N_n^\alpha) &= \mu_2^\beta(T, P, N_1^\beta, N_2^\beta, \dots, N_n^\beta) \\ &\dots = \dots \\ \mu_n^\alpha(T, P, N_1^\alpha, N_2^\alpha, \dots, N_n^\alpha) &= \mu_n^\beta(T, P, N_1^\beta, N_2^\beta, \dots, N_n^\beta)\end{aligned}$$

These are n equalities involving $2n$ μ 's. If we had 3 phases, we would have $2n$ equalities for $3n$ μ 's ($3n$ possible equalities, but only $2n$ independent ones), for 4 phases, $3n$ (independent) equalities, etc. **For n species in φ phases there are $n(\varphi - 1)$ independent chemical-potential equalities that express material equilibrium.** (This will be used later.)

5.1. Material equilibrium

Phase Equilibria

Multicomponent systems

Now, since the chemical potential is intensive, it should be expressible as a function of intensive and specific variables only. For this reason, it is convenient to introduce the *molar fraction* of chemical i in each phase, e.g., α , as $x_i^\alpha \equiv N_i^\alpha / N^\alpha$, where

$$N^\alpha \equiv \sum_{i=1}^n N_i^\alpha.$$

By dividing both sides by N^α , we get:

$$1 = \sum_{i=1}^n x_i^\alpha.$$

Ergo, in a system with n components, only $n - 1$ molar fractions are needed to describe the composition. Therefore, **for n species in φ phases, $\varphi(n - 1)$ molar fractions are needed in total.** (This will also be used later.) In sum, for each species i in a phase α :

$$\mu_i^\alpha = \mu_i^\alpha(T, P, x_1^\alpha, x_2^\alpha, \dots, x_{n-1}^\alpha)$$

5.1. Material equilibrium

Phase Equilibria

Multicomponent systems

So, returning to the case of n species in *two* coexisting phases, α and β , in material equilibrium at T and P , the earlier set of equations can be expressed as:

$$\begin{aligned}\mu_1^\alpha(T, P, x_1^\alpha, x_2^\alpha, \dots, x_{n-1}^\alpha) &= \mu_1^\beta(T, P, x_1^\beta, x_2^\beta, \dots, x_{n-1}^\beta) \\ \mu_2^\alpha(T, P, x_1^\alpha, x_2^\alpha, \dots, x_{n-1}^\alpha) &= \mu_2^\beta(T, P, x_1^\beta, x_2^\beta, \dots, x_{n-1}^\beta) \\ &\dots = \dots \\ \mu_n^\alpha(T, P, x_1^\alpha, x_2^\alpha, \dots, x_{n-1}^\alpha) &= \mu_n^\beta(T, P, x_1^\beta, x_2^\beta, \dots, x_{n-1}^\beta)\end{aligned}$$

Note that if we fix T and P , only for the case of two components ($n = 2$) the number of unknowns (x_1^α, x_1^β) equals the number of equations (2), and we can solve them to find the concentrations of each component. In other words, when two components are present in two coexisting phases (e.g., liquid-gas), their compositions are fixed when T and P are given. In the case of $n = 2$ and $\varphi = 3$ phases, we are not free to simultaneously specify T and P , but only one of them (5 unknowns, 4 equations). And 4 phases would only coexist at a specific T , P and composition. [2]

5.1. Material equilibrium

Thermodynamic degrees of freedom

Gibbs' phase rule [2]

The number of independent *intensive* variables in a system is called **thermodynamic degrees of freedom**, f . **Gibbs** announced a (now-famous) relationship that determines the f for n -component systems as a function of the number of coexisting phases, φ :

$$f = 2 + n - \varphi \quad (2)$$

f represents the number of intensive variables that can be independently assigned in the coexistence zone. It must equal the total number of parameters needed to characterize the system (T , P , and the $n - 1$ molar fractions in φ phases), namely $2 + \varphi(n - 1)$, minus the number of chemical potential equalities that constrain their values, namely $n(\varphi - 1)$.

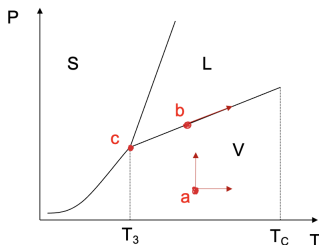
f can also be obtained directly from the intensive variables: In the presence of n components we would have $2 + n$ intensive variables: T , P , μ_1 , ..., μ_n , all being the same in the coexisting phases. However, for each phase there is a Gibbs-Duhem relationship that constrains their variation. Such φ Gibbs-Duhem equations reduce the number of independent variables to $2 + n$ minus φ ; thus the Gibbs rule given above.

5.1. Material equilibrium

Thermodynamic degrees of freedom

Degrees of freedom in a monocomponent system

If we consider the P - T diagram of a closed system with one component ($n = 1$), the number of degrees of freedom of the system (1 phase) is $f = 2$ and the state of the system can be represented in a plane like the one in the figure,



- ▶ Point **a** (single phase) can move on the P - T plane in two directions and without phase change, until reaching the coexistence line.
- ▶ Point **b** (two coexisting phases) can only move on the curve $P = P^{(LV)}(T)$, as $f = 2 + 1 - 2 = 1$ the system has only one degree of freedom. The two phases coexist on a line.
- ▶ Point **c** (three coexisting phases) only occurs at a specific point, the **triple point**, and one cannot move away from it without losing one of the phases. The system has 0 degrees of freedom, since $f = 2 + 1 - 3 = 0$.

5.1. Material equilibrium

Clausius-Clapeyron equation

The shape of each coexistence curve in the P - T phase diagram are determined by the properties of the substance in question. **Clapeyron** found that the slope satisfies a differential equation of the form,

$$\frac{dP}{dT} = \frac{\Delta H_v}{T \Delta v}$$

where ΔH_v is the *molar* latent heat of vaporization (similarly, for melting or sublimation) and Δv is the molar volume change between the phases. This allows us to reconstruct the coexistence curve from measurable quantities.

In the cases of liquid-vapor or solid-vapor coexistence, some approximations can be made. The most important one is: $\Delta v = v_{vap} - v_{liq} \simeq v_{vap}$. In addition, under conditions of low pressure and moderately high temperature, the gas will behave similarly to an ideal gas, so we can approximately write $v_{vap} = RT/P$. Then, if over the considered range we can take $\Delta H_v \simeq \text{constant}$:

$$\frac{dP}{dT} = \frac{P \Delta H_v}{RT^2} \Rightarrow \frac{dP}{P} = \Delta H_v \frac{dT}{RT^2};$$

$$\ln \frac{P_2}{P_1} = -\Delta H_v \left(\frac{1}{RT_2} - \frac{1}{RT_1} \right)$$

This is known as the **Claussius-Clapeyron equation**:

5.1. Material equilibrium

Clausius-Clapeyron equation

Proof of the Clapeyron equation

Consider a point along the coexistence curve at (T, P) and a differential displacement to $(T + dT, P + dP)$. At each point, the chemical potentials of both phases are equal, specifically, $\mu^\alpha(T + dT, P + dP) = \mu^\beta(T + dT, P + dP)$. By expanding each side by a Taylor series around (T, P) , we get, to first order,

$$\mu^\alpha(T, P) + \left. \frac{\partial \mu^\alpha}{\partial T} \right|_P dT + \left. \frac{\partial \mu^\alpha}{\partial P} \right|_T dP = \mu^\beta(T, P) + \left. \frac{\partial \mu^\beta}{\partial T} \right|_P dT + \left. \frac{\partial \mu^\beta}{\partial P} \right|_T dP$$

Now, from the Gibbs-Duhem equation, $d\mu = -s dT + v dP$, we see that $\left. \frac{\partial \mu^\alpha}{\partial T} \right|_P = -s^\alpha$ and $\left. \frac{\partial \mu^\alpha}{\partial P} \right|_T = v^\alpha$. Substituting these in the above, and after cancelling the zero-order terms since $\mu^\alpha(T, P) = \mu^\beta(T, P)$, we get:

$$-(s^\alpha - s^\beta)dT + (v^\alpha - v^\beta)dP = 0$$

$$\frac{dP}{dT} = \frac{\Delta s|_{T,P}}{\Delta v|_{T,P}} = \frac{\Delta H}{T \Delta v}$$

Recommended reading

- [1] I. Levine, *Principios de Fisicoquímica*, McGraw-Hill; Chap. 4
- [2] H.B. Callen, *Termodinámica*, Ediciones AC; Chap. 9.
- [3] https://ca.wikipedia.org/wiki/Hermann_von_Helmholtz
- [4] <http://rbx-exhibit2000.scs.illinois.edu/vanderwaals.htm>
- [5] https://ca.wikipedia.org/wiki/James_Clerk_Maxwell
- [6] <https://www.math24.net/van-der-waals-equation>